

Trajectory_Analysis_Perturbations

June 1, 2026

```
[ ]: !pip install scanpy
      !pip install anndata
      !pip3 install igraph
      !pip install celltypist
      !pip install decoupler
      !pip install fa2-modified
      !pip install louvain
      !pip install perpy
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Requirement already satisfied: scanpy in /usr/local/lib/python3.12/dist-packages
(1.11.5)
Requirement already satisfied: anndata>=0.8 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages
(from scanpy) (1.5.2)
Requirement already satisfied: legacy-api-wrap>=1.4.1 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (1.4.1)
Requirement already satisfied: matplotlib>=3.7.5 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (3.10.0)
Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: networkx>=2.7.1 in
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Requirement already satisfied: numba!=0.62.0rc1,>=0.57.1 in
/usr/local/lib/python3.12/dist-packages (from scanpy) (0.60.0)
Requirement already satisfied: numpy>=1.24.1 in /usr/local/lib/python3.12/dist-
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Requirement already satisfied: pandas>=1.5.3 in /usr/local/lib/python3.12/dist-
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 Requirement already satisfied: statsmodels>=0.14.5 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.14.5)
 Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from scanpy) (4.67.1)
 Requirement already satisfied: typing-extensions in /usr/local/lib/python3.12/dist-packages (from scanpy) (4.15.0)
 Requirement already satisfied: umap-learn>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from scanpy) (0.5.9.post2)
 Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy) (1.12.0)
 Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy) (3.1.3)
 Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.3.3)
 Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (4.60.1)
 Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (1.4.9)
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 Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (3.2.5)
 Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy) (2.9.0.post0)
 Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba!=0.62.0rc1,>=0.57.1->scanpy) (0.43.0)
 Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5.3->scanpy) (2025.2)
 Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.5.3->scanpy) (2025.2)
 Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.1.3->scanpy) (3.6.0)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.7.5->scanpy) (1.17.0)
 Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-

packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy) (0.8.1.post1)
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 numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy) (0.16.3)
 Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas!=2.1.2,>=2.1.0->anndata) (1.17.0)

Requirement already satisfied: igraph in /usr/local/lib/python3.12/dist-packages (0.11.9)

Requirement already satisfied: texttable>=1.6.2 in /usr/local/lib/python3.12/dist-packages (from igraph) (1.7.0)

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Requirement already satisfied: openpyxl>=3.0.4 in /usr/local/lib/python3.12/dist-packages (from celltypist) (3.1.5)

Requirement already satisfied: click>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from celltypist) (8.3.0)

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Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl>=3.0.4->celltypist) (2.0.0)

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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->celltypist) (2025.10.5)

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 Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.1->celltypist) (3.6.0)
 Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy>=1.7.0->celltypist) (1.12.0)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.8->scanpy>=1.7.0->celltypist) (3.1.3)

Requirement already satisfied: texttable>=1.6.2 in /usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->leidenalg>=0.9.0->celltypist) (1.7.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.3.3)

Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (1.4.9)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.7.5->scanpy>=1.7.0->celltypist) (3.2.5)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from numba!=0.62.0rc1,>=0.57.1->scanpy>=1.7.0->celltypist) (0.43.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.0.5->celltypist) (1.17.0)

Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.8.1.post1)

Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.8->scanpy>=1.7.0->celltypist) (0.16.3)

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Requirement already satisfied: decoupler in /usr/local/lib/python3.12/dist-packages (2.1.1)

Requirement already satisfied: adjusttext in /usr/local/lib/python3.12/dist-packages (from decoupler) (1.3.0)

Requirement already satisfied: anndata in /usr/local/lib/python3.12/dist-packages (from decoupler) (0.12.4)

Requirement already satisfied: docrep in /usr/local/lib/python3.12/dist-packages (from decoupler) (0.3.2)

Requirement already satisfied: marsilea in /usr/local/lib/python3.12/dist-packages (from decoupler) (0.5.6)
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (1.4.9)

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Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->decoupler) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas!=2.1.2,>=2.1.0->anndata->decoupler) (2025.2)

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Requirement already satisfied: pyarrow>=10.0.1 in /usr/local/lib/python3.12/dist-packages (from pandas[parquet]->marsilea->decoupler) (18.1.0)

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from donfig>=0.8->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (6.0.3)

Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata->decoupler) (2.8)

Requirement already satisfied: fa2-modified in /usr/local/lib/python3.12/dist-packages (0.4)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from fa2-modified) (2.0.2)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from fa2-modified) (1.15.3)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from fa2-modified) (4.67.1)

Requirement already satisfied: louvain in /usr/local/lib/python3.12/dist-

packages (0.8.2)
 Requirement already satisfied: igraph<0.12,>=0.10.0 in
 /usr/local/lib/python3.12/dist-packages (from louvain) (0.11.9)
 Requirement already satisfied: texttable>=1.6.2 in
 /usr/local/lib/python3.12/dist-packages (from igraph<0.12,>=0.10.0->louvain)
 (1.7.0)
 Requirement already satisfied: perpty in /usr/local/lib/python3.12/dist-packages
 (1.0.3)
 Requirement already satisfied: adjusttext in /usr/local/lib/python3.12/dist-
 packages (from perpty) (1.3.0)
 Requirement already satisfied: arviz in /usr/local/lib/python3.12/dist-packages
 (from perpty) (0.22.0)
 Requirement already satisfied: blitzgsea in /usr/local/lib/python3.12/dist-
 packages (from perpty) (1.3.54)
 Requirement already satisfied: fast-array-utils[accel,sparse] in
 /usr/local/lib/python3.12/dist-packages (from perpty) (1.2.5)
 Requirement already satisfied: flax in /usr/local/lib/python3.12/dist-packages
 (from perpty) (0.10.7)
 Requirement already satisfied: funsor in /usr/local/lib/python3.12/dist-packages
 (from perpty) (0.4.5)
 Requirement already satisfied: lamin-utils in /usr/local/lib/python3.12/dist-
 packages (from perpty) (0.15.0)
 Requirement already satisfied: mudata in /usr/local/lib/python3.12/dist-packages
 (from perpty) (0.3.2)
 Requirement already satisfied: numpyro in /usr/local/lib/python3.12/dist-
 packages (from perpty) (0.19.0)
 Requirement already satisfied: openpyxl in /usr/local/lib/python3.12/dist-
 packages (from perpty) (3.1.5)
 Requirement already satisfied: ott-jax in /usr/local/lib/python3.12/dist-
 packages (from perpty) (0.5.2)
 Requirement already satisfied: pubchempy in /usr/local/lib/python3.12/dist-
 packages (from perpty) (1.0.5)
 Requirement already satisfied: pyarrow in /usr/local/lib/python3.12/dist-
 packages (from perpty) (18.1.0)
 Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
 packages (from perpty) (2.32.4)
 Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages
 (from perpty) (13.9.4)
 Requirement already satisfied: scanpy in /usr/local/lib/python3.12/dist-packages
 (from perpty) (1.11.5)
 Requirement already satisfied: scikit-learn>=1.4 in
 /usr/local/lib/python3.12/dist-packages (from perpty) (1.6.1)
 Requirement already satisfied: scikit-misc in /usr/local/lib/python3.12/dist-
 packages (from perpty) (0.5.1)
 Requirement already satisfied: sparsecca in /usr/local/lib/python3.12/dist-
 packages (from perpty) (0.3.1)
 Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.12/dist-
 packages (from scikit-learn>=1.4->perpty) (2.0.2)

Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.4->pertpy) (1.15.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.4->pertpy) (1.5.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=1.4->pertpy) (3.6.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from adjusttext->pertpy) (3.10.0)
Requirement already satisfied: setuptools>=60.0.0 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (75.2.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (25.0)
Requirement already satisfied: pandas>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (2.2.2)
Requirement already satisfied: xarray>=2023.7.0 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (2025.10.1)
Requirement already satisfied: h5netcdf>=1.0.2 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (1.7.2)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (4.15.0)
Requirement already satisfied: xarray-einstats>=0.3 in /usr/local/lib/python3.12/dist-packages (from arviz->pertpy) (0.9.1)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from blitzgsea->pertpy) (4.67.1)
Requirement already satisfied: statsmodels in /usr/local/lib/python3.12/dist-packages (from blitzgsea->pertpy) (0.14.5)
Requirement already satisfied: mpmath in /usr/local/lib/python3.12/dist-packages (from blitzgsea->pertpy) (1.3.0)
Requirement already satisfied: numba>=0.57 in /usr/local/lib/python3.12/dist-packages (from fast-array-utils[accel,sparse]->pertpy) (0.60.0)
Requirement already satisfied: jax>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.7.2)
Requirement already satisfied: msgpack in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (1.1.2)
Requirement already satisfied: optax in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.2.6)
Requirement already satisfied: orbax-checkpoint in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.11.25)
Requirement already satisfied: tensorstore in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.1.78)
Requirement already satisfied: PyYAML>=5.4.1 in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (6.0.3)
Requirement already satisfied: treescope>=0.1.7 in /usr/local/lib/python3.12/dist-packages (from flax->pertpy) (0.1.10)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->pertpy) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->pertpy) (2.19.2)

Requirement already satisfied: makefun in /usr/local/lib/python3.12/dist-packages (from funsor->pertpy) (1.16.0)

Requirement already satisfied: multipledispatch in /usr/local/lib/python3.12/dist-packages (from funsor->pertpy) (1.0.0)

Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.12/dist-packages (from funsor->pertpy) (3.4.0)

Requirement already satisfied: anndata>=0.10.8 in /usr/local/lib/python3.12/dist-packages (from mudata->pertpy) (0.12.4)

Requirement already satisfied: jaxlib>=0.4.25 in /usr/local/lib/python3.12/dist-packages (from numpyro->pertpy) (0.7.2)

Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl->pertpy) (2.0.0)

Requirement already satisfied: jaxopt>=0.8 in /usr/local/lib/python3.12/dist-packages (from ott-jax->pertpy) (0.8.5)

Requirement already satisfied: lineax>=0.0.7 in /usr/local/lib/python3.12/dist-packages (from ott-jax->pertpy) (0.0.8)

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->pertpy) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->pertpy) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->pertpy) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->pertpy) (2025.10.5)

Requirement already satisfied: h5py>=3.7.0 in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (3.15.1)

Requirement already satisfied: legacy-api-wrap>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (1.4.1)

Requirement already satisfied: natsort in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (8.4.0)

Requirement already satisfied: networkx>=2.7.1 in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (3.5)

Requirement already satisfied: patsy!=1.0.0 in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (1.0.2)

Requirement already satisfied: pynndescent>=0.5.13 in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (0.5.13)

Requirement already satisfied: seaborn>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (0.13.2)

Requirement already satisfied: session-info2 in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (0.2.3)

Requirement already satisfied: umap-learn>=0.5.6 in /usr/local/lib/python3.12/dist-packages (from scanpy->pertpy) (0.5.9.post2)

Requirement already satisfied: pyomo in /usr/local/lib/python3.12/dist-packages (from sparsecca->pertpy) (6.9.5)

Requirement already satisfied: array-api-compat>=1.7.1 in /usr/local/lib/python3.12/dist-packages (from anndata>=0.10.8->mudata->pertpy) (1.12.0)

Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in

/usr/local/lib/python3.12/dist-packages (from anndata>=0.10.8->mudata->pertpy)
 (3.1.3)
 Requirement already satisfied: ml_dtypes>=0.5.0 in
 /usr/local/lib/python3.12/dist-packages (from jax>=0.6.0->flax->pertpy) (0.5.3)
 Requirement already satisfied: equinox>=0.11.10 in
 /usr/local/lib/python3.12/dist-packages (from lineax>=0.0.7->ott-jax->pertpy)
 (0.13.2)
 Requirement already satisfied: jaxtyping>=0.2.24 in
 /usr/local/lib/python3.12/dist-packages (from lineax>=0.0.7->ott-jax->pertpy)
 (0.3.3)
 Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-
 packages (from markdown-it-py>=2.2.0->rich->pertpy) (0.1.2)
 Requirement already satisfied: contourpy>=1.0.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy)
 (1.3.3)
 Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-
 packages (from matplotlib->adjusttext->pertpy) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy)
 (4.60.1)
 Requirement already satisfied: kiwisolver>=1.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy)
 (1.4.9)
 Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
 packages (from matplotlib->adjusttext->pertpy) (11.3.0)
 Requirement already satisfied: pyparsing>=2.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy)
 (3.2.5)
 Requirement already satisfied: python-dateutil>=2.7 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib->adjusttext->pertpy)
 (2.9.0.post0)
 Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in
 /usr/local/lib/python3.12/dist-packages (from numba>=0.57->fast-array-
 utils[accel,sparse]->pertpy) (0.43.0)
 Requirement already satisfied: absl-py>=0.7.1 in /usr/local/lib/python3.12/dist-
 packages (from optax->flax->pertpy) (1.4.0)
 Requirement already satisfied: chex>=0.1.87 in /usr/local/lib/python3.12/dist-
 packages (from optax->flax->pertpy) (0.1.90)
 Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
 packages (from pandas>=2.1.0->arviz->pertpy) (2025.2)
 Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
 packages (from pandas>=2.1.0->arviz->pertpy) (2025.2)
 Requirement already satisfied: etils[epath,epy] in
 /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy)
 (1.13.0)
 Requirement already satisfied: nest_asyncio in /usr/local/lib/python3.12/dist-
 packages (from orbax-checkpoint->flax->pertpy) (1.6.0)
 Requirement already satisfied: aiofiles in /usr/local/lib/python3.12/dist-

packages (from orbax-checkpoint->flax->pertpy) (24.1.0)
 Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (5.29.5)
 Requirement already satisfied: humanize in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (4.14.0)
 Requirement already satisfied: simplejson>=3.16.0 in /usr/local/lib/python3.12/dist-packages (from orbax-checkpoint->flax->pertpy) (3.20.2)
 Requirement already satisfied: ply in /usr/local/lib/python3.12/dist-packages (from pyomo->sparsecca->pertpy) (3.11)
 Requirement already satisfied: toolz>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from chex>=0.1.87->optax->flax->pertpy) (0.12.1)
 Requirement already satisfied: wadler-lindig>=0.1.0 in /usr/local/lib/python3.12/dist-packages (from equinox>=0.11.10->lineax>=0.0.7->ott-jax->pertpy) (0.1.7)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->adjusttext->pertpy) (1.17.0)
 Requirement already satisfied: donfig>=0.8 in /usr/local/lib/python3.12/dist-packages (from zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->mudata->pertpy) (0.8.1.post1)
 Requirement already satisfied: numcodecs>=0.14 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->mudata->pertpy) (0.16.3)
 Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from etils[epath,epy]->orbax-checkpoint->flax->pertpy) (2025.3.0)
 Requirement already satisfied: importlib_resources in /usr/local/lib/python3.12/dist-packages (from etils[epath,epy]->orbax-checkpoint->flax->pertpy) (6.5.2)
 Requirement already satisfied: zipp in /usr/local/lib/python3.12/dist-packages (from etils[epath,epy]->orbax-checkpoint->flax->pertpy) (3.23.0)
 Requirement already satisfied: crc32c>=2.7 in /usr/local/lib/python3.12/dist-packages (from numcodecs[crc32c]>=0.14->zarr!=3.0.*,>=2.18.7->anndata>=0.10.8->mudata->pertpy) (2.8)

```
[ ]: #Import core single cell datasets
```

```
import scanpy as sc
import anndata as ad
import numpy as np
import pertpy as pt
```

```
[ ]: #!wget https://datasets.cellxgene.cziscience.com/
↳ fdf57c52-ad71-4004-9db2-a962e849b524.h5ad
!wget https://datasets.cellxgene.cziscience.com/
↳ 497ab773-4fd5-4263-9fdf-7b42a68f1351.h5ad
```

```
--2025-10-28 16:03:05-- https://datasets.cellxgene.cziscience.com/497ab773-4fd5-4263-9fdf-7b42a68f1351.h5ad
Resolving datasets.cellxgene.cziscience.com
(datasets.cellxgene.cziscience.com)... 13.35.185.126, 13.35.185.84, 13.35.185.112, ...
Connecting to datasets.cellxgene.cziscience.com
(datasets.cellxgene.cziscience.com)|13.35.185.126|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 74549927 (71M) [binary/octet-stream]
Saving to: '497ab773-4fd5-4263-9fdf-7b42a68f1351.h5ad.1'
```

```
497ab773-4fd5-4263- 100%[=====>] 71.10M 97.2MB/s in 0.7s
```

```
2025-10-28 16:03:06 (97.2 MB/s) - '497ab773-4fd5-4263-9fdf-7b42a68f1351.h5ad.1'
saved [74549927/74549927]
```

```
[ ]: bm_mtx = sc.read_h5ad('/content/497ab773-4fd5-4263-9fdf-7b42a68f1351.h5ad')
bm_mtx.var_names_make_unique()
```

```
[ ]: bm_mtx.shape
```

```
[ ]: (14783, 17374)
```

```
[ ]: # A useful step for older datasets
bm_mtx.var_names_make_unique()
bm_mtx.obs_names_make_unique()
```

```
[ ]: bm_mtx.var.head()
```

```
[ ]:
      n_cells  feature_is_filtered  feature_name  feature_reference \
ENSG00000161920      927           False      MED11  NCBITaxon:9606
ENSG00000122335      429           False      SERAC1  NCBITaxon:9606
ENSG00000175548      149           False      ALG10B  NCBITaxon:9606
ENSG00000100330      426           False      MTMR3  NCBITaxon:9606
ENSG00000176340     7171           False      COX8A  NCBITaxon:9606

      feature_biotype  feature_length  feature_type
ENSG00000161920      gene           754  protein_coding
ENSG00000122335      gene          2683  protein_coding
ENSG00000175548      gene          1773  protein_coding
ENSG00000100330      gene           634  protein_coding
ENSG00000176340      gene           494  protein_coding
```

```
[ ]: bm_mtx.shape
```

```
[ ]: (14783, 17374)
```

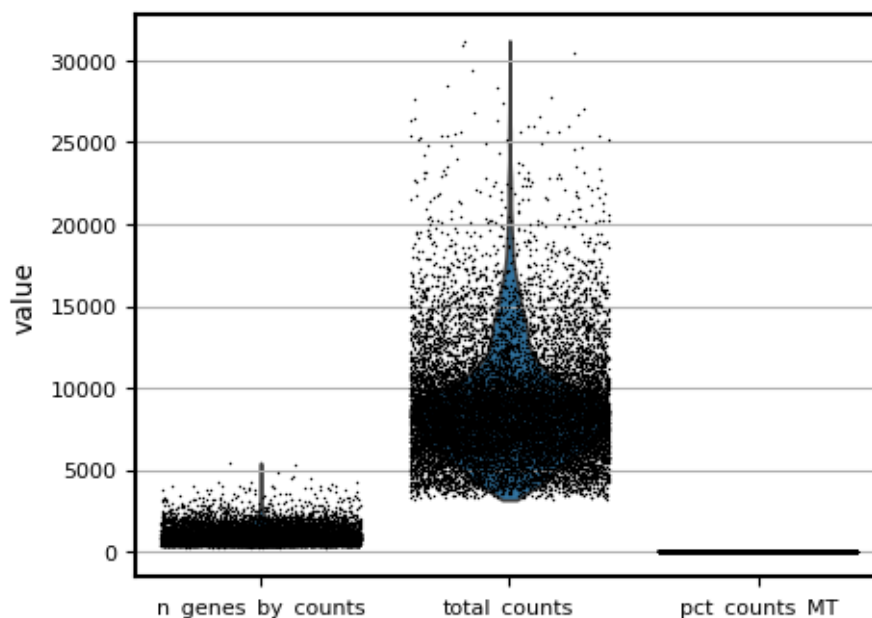
```
[ ]: bm_mtx.var['MT'] = bm_mtx.var_names.str.startswith("MT-")
bm_mtx.var['RIBO'] = bm_mtx.var_names.str.startswith("RPS", "RPL")
bm_mtx.var['HB'] = bm_mtx.var_names.str.startswith("^HB[^P]")

[ ]: sc.pp.calculate_qc_metrics(
    bm_mtx, qc_vars=["MT", 'RIBO', 'HB'], inplace=True, log1p=True
)

[ ]: import matplotlib.pyplot as plt

plt.rcParams["figure.figsize"] = (5,4) # Adjust figure size
plt.rcParams["axes.grid"] = True # Add grid to plots
plt.rcParams["axes.edgecolor"] = "black" # Set plot border color
plt.rcParams["axes.linewidth"] = 1.5 # Set plot border width
plt.rcParams["axes.facecolor"] = "white" # Set background color
plt.rcParams["axes.labelcolor"] = "black" # Set label color
plt.rcParams["xtick.color"] = "black" # Set x-axis tick color
plt.rcParams["ytick.color"] = "black" # Set y-axis tick color
plt.rcParams["text.color"] = "black" # Set text color
%matplotlib inline

[ ]: sc.pl.violin(
    bm_mtx,
    ["n_genes_by_counts", 'total_counts', 'pct_counts_MT'],
    jitter=0.4,
    multi_panel=False,
)
```



```
[ ]: bm_mtx.obs_keys()
```

```
/tmp/ipython-input-3505303993.py:1: FutureWarning: Use obs (e.g. `k` in  
adata.obs` or `str(adata.obs.columns.tolist())`) instead of AnnData.obs_keys,  
AnnData.obs_keys is deprecated and will be removed in the future.
```

```
bm_mtx.obs_keys()
```

```
[ ]: ['disease stage',  
      'treatment',  
      'timepoint',  
      'Dataset',  
      'sample',  
      'disease_original',  
      'disease_general',  
      'COVID-19 Condition',  
      'Lineage',  
      'Cell.group',  
      'Cell.class_reannotated',  
      'n_genes',  
      'n_counts',  
      'percent_mito',  
      'tissue_original',  
      'tissue_ontology_term_id',  
      'disease_ontology_term_id',  
      'donor_id',  
      'development_stage_ontology_term_id',  
      'assay_ontology_term_id',  
      'cell_type_ontology_term_id',  
      'self_reported_ethnicity_ontology_term_id',  
      'sex_ontology_term_id',  
      'is_primary_data',  
      'suspension_type',  
      'tissue_type',  
      'cell_type',  
      'assay',  
      'disease',  
      'sex',  
      'tissue',  
      'self_reported_ethnicity',  
      'development_stage',  
      'observation_joinid',  
      'n_genes_by_counts',  
      'log1p_n_genes_by_counts',  
      'total_counts',  
      'log1p_total_counts',  
      'pct_counts_in_top_50_genes',
```

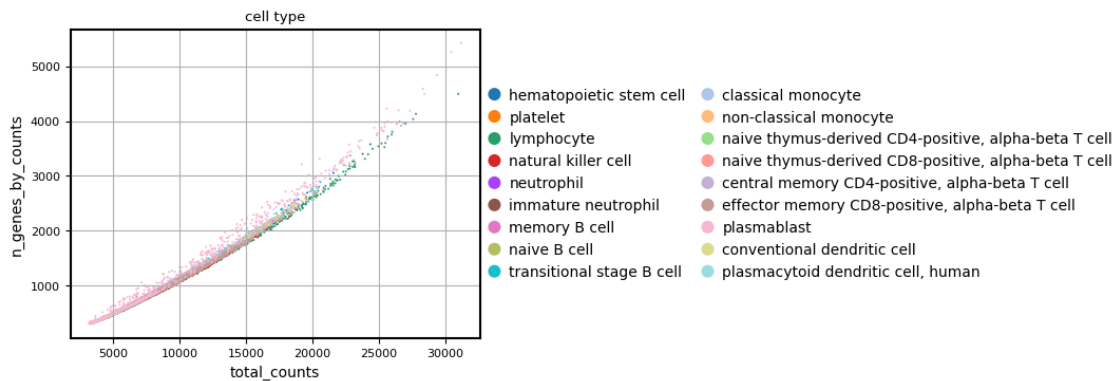


```

'pct_counts_in_top_100_genes',
'pct_counts_in_top_200_genes',
'pct_counts_in_top_500_genes',
'total_counts_MT',
'log1p_total_counts_MT',
'pct_counts_MT',
'total_counts_RIBO',
'log1p_total_counts_RIBO',
'pct_counts_RIBO',
'total_counts_HB',
'log1p_total_counts_HB',
'pct_counts_HB']

```

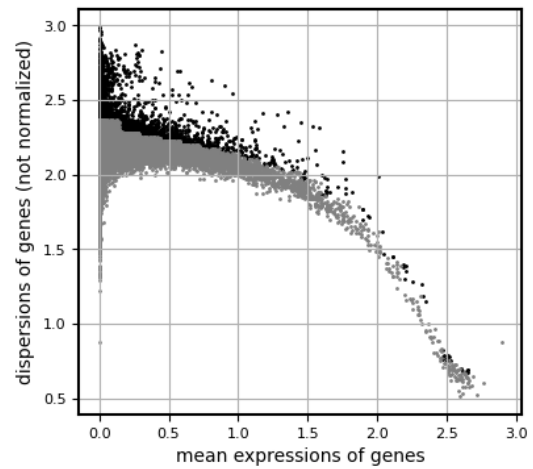
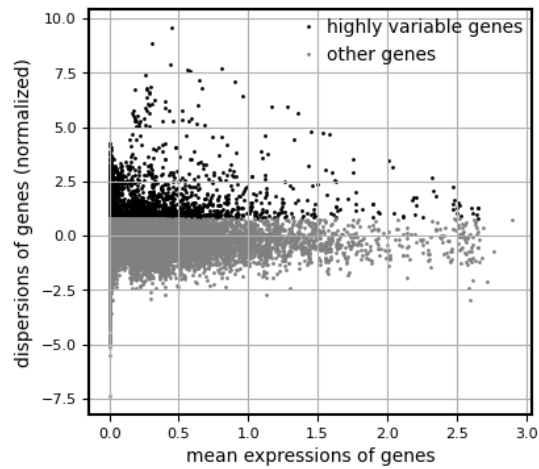
```
[ ]: sc.pl.scatter(bm_mtx, "total_counts", "n_genes_by_counts", color="cell_type")
```



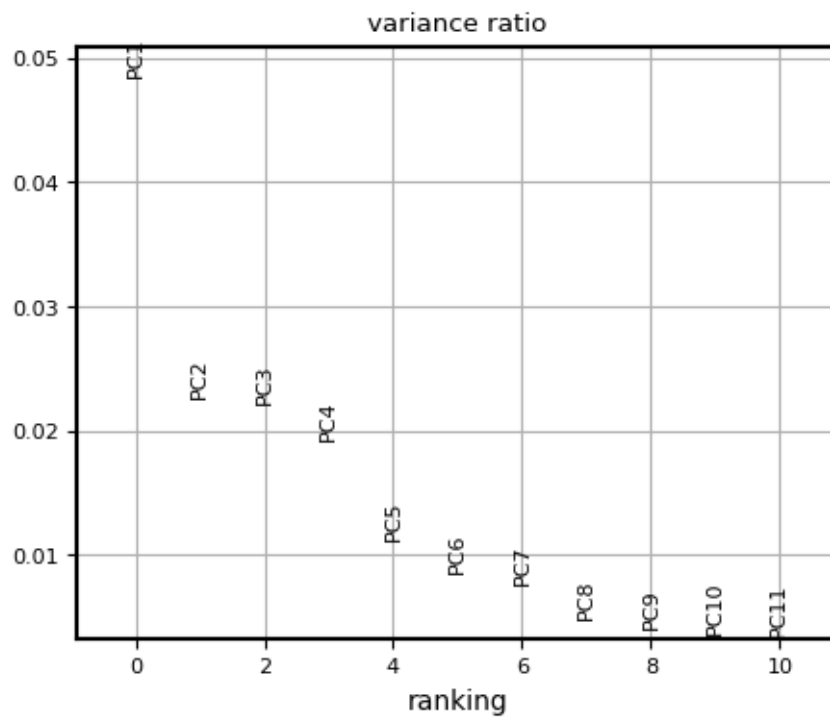
```
[ ]: #sc.pp.scrublet(bm_mtx)
```

```
[ ]: #Normalisation
bm_mtx.layers["counts"] = bm_mtx.X.copy()
sc.pp.normalize_total(bm_mtx)
sc.pp.log1p(bm_mtx)
```

```
[ ]: #Feature selection
sc.pp.highly_variable_genes(bm_mtx, n_top_genes=2500)
sc.pl.highly_variable_genes(bm_mtx)
```



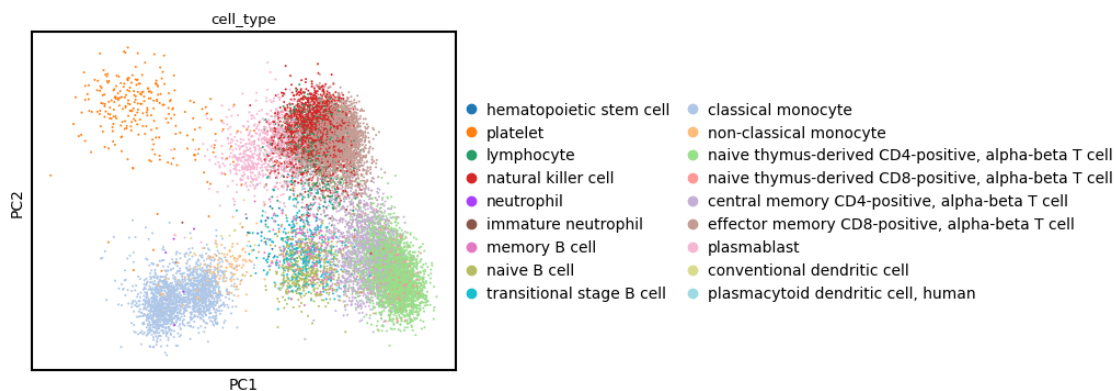
```
[ ]: #Dim Reduction
sc.tl.pca(bm_mtx)
sc.pl.pca_variance_ratio(bm_mtx, n_pcs=10, log=False)
```



```
[ ]: bm_mtx.var_names
```

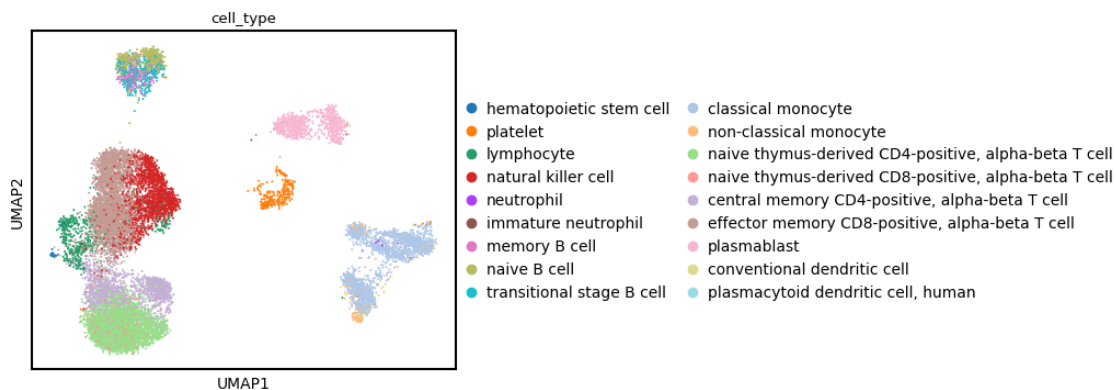
```
[ ]: Index(['ENSG00000161920', 'ENSG00000122335', 'ENSG00000175548',
          'ENSG00000100330', 'ENSG00000176340', 'ENSG00000179846',
          'ENSG00000204860', 'ENSG00000172260', 'ENSG00000141424',
          'ENSG00000164512',
          ...
          'ENSG00000164114', 'ENSG00000151702', 'ENSG00000224578',
          'ENSG00000138756', 'ENSG00000111052', 'ENSG00000176946',
          'ENSG00000150456', 'ENSG00000284934', 'ENSG00000261842',
          'ENSG00000260456'],
          dtype='object', length=17374)
```

```
[ ]: sc.pl.pca(bm_mtx, color=["cell_type"], cmap="coolwarm")
```



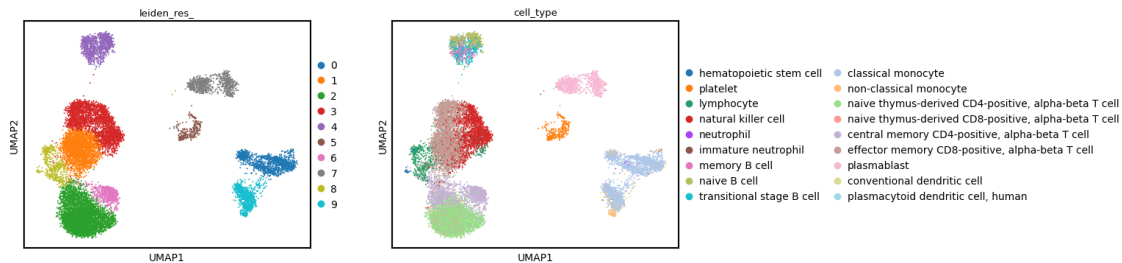
```
[ ]: sc.pp.neighbors(bm_mtx)
      sc.tl.umap(bm_mtx)
```

```
[ ]: sc.pl.umap(
      bm_mtx,
      color=["cell_type"],
      size=8,
      )
```



```
[ ]: sc.tl.leiden(bm_mtx, flavor="igraph", n_iterations=10, key_added="leiden_res_",
↳ resolution=0.5 )
```

```
[ ]: sc.pl.umap(
    bm_mtx,
    color=["leiden_res_", 'cell_type'],
    size=8,
)
```



```
[ ]: import decoupler as dc
```

```
[ ]: !wget wget -O result.txt 'http://www.ensembl.org/biomart/martservice?query=?
↳ xml version="1.0" encoding="UTF-8"?><!DOCTYPE Query><Query
↳ virtualSchemaName = "default" formatter = "CSV" header = "0" uniqueRows =
↳ "0" count = "" datasetConfigVersion = "0.6" ><Dataset name =
↳ "hsapiens_gene_ensembl" interface = "default" ><Attribute name =
↳ "ensembl_gene_id" /><Attribute name = "external_gene_name" /></Dataset></
↳ Query>'
```

```
--2025-10-28 16:03:51-- http://wget/
Resolving wget (wget)... failed: Name or service not known.
wget: unable to resolve host address 'wget'
--2025-10-28 16:03:51-- http://www.ensembl.org/biomart/martservice?query=%3C?xm
l%20version=%221.0%22%20encoding=%22UTF-
8%22%3E%3C!DOCTYPE%20Query%3E%3CQuery%20%20virtualSchemaName%20=%20%22default%2
2%20formatter%20=%20%22CSV%22%20header%20=%20%220%22%20uniqueRows%20=%20%220%22%
20count%20=%20%22%22%20datasetConfigVersion%20=%20%220.6%22%20%3E%3CDataset%20na
me%20=%20%22hsapiens_gene_ensembl%22%20interface%20=%20%22default%22%20%3E%3CAtt
ribute%20name%20=%20%22ensembl_gene_id%22%20/%3E%3CAttribute%20name%20=%20%22ext
ernal_gene_name%22%20/%3E%3C/Dataset%3E%3C/Query%3E
Resolving www.ensembl.org (www.ensembl.org)... 193.62.193.83
Connecting to www.ensembl.org (www.ensembl.org)|193.62.193.83|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/plain]
```

Saving to: 'result.txt'

result.txt [<=>] 1.71M 808KB/s in 2.2s

2025-10-28 16:03:54 (808 KB/s) - 'result.txt' saved [1795328]

FINISHED --2025-10-28 16:03:54--

Total wall clock time: 3.1s

Downloaded: 1 files, 1.7M in 2.2s (808 KB/s)

```
[ ]: import pandas as pd

ensembl_var = pd.read_csv('/content/result.txt', header = None)

ensembl_var.columns = ['ensembl_gene_id', 'gene_name']

ensembl_var.head(3)
```

```
[ ]:      ensembl_gene_id gene_name
0  ENSG00000210049      MT-TF
1  ENSG00000211459      MT-RNR1
2  ENSG00000210077      MT-TV
```

```
[ ]: # Query Omnipath and get PanglaoDB
markers = dc.op.resource(name="PanglaoDB", organism="human")

# Keep canonical cell type markers alone
markers = markers[markers["canonical_marker"]]

# Remove duplicated entries
markers = markers[~markers.duplicated(["cell_type", "genesymbol"])]

#Format because dc only accepts cell_type and genesymbol

markers = markers.rename(columns={"cell_type": "source", "genesymbol": "target"})
markers = markers[["source", "target"]]

markers.head()
```

```
[ ]:      source target
2      Bergmann glia      A2M
4      Mast cells      A4GALT
5  Embryonic stem cells      A7KBS4
8      Pinealocytes      AANAT
```

```
[ ]: #correct target to ensemble
markers = markers.merge(ensembl_var, left_on="target", right_on="gene_name",
    how="left")
markers = markers.drop(columns=["target"])
# Remove duplicated entries
markers = markers[~markers.duplicated(["source", "ensembl_gene_id"])]

#Format because dc only accepts cell_type and genesymbol
markers = markers.rename(columns={"source": "source", "ensembl_gene_id":
    "target"})

markers = markers[["source", "target"]]
markers = markers.dropna()

markers.head()
```

```
[ ]:
      source      target
0  Bergmann glia  ENSG00000175899
1    Mast cells  ENSG00000128274
3  Pinealocytes  ENSG00000129673
4    Astrocytes  ENSG00000129673
5  Sertoli cells  ENSG00000205002
```

```
[ ]: bm_mtx.var_names
```

```
[ ]: Index(['ENSG00000161920', 'ENSG00000122335', 'ENSG00000175548',
      'ENSG00000100330', 'ENSG00000176340', 'ENSG00000179846',
      'ENSG00000204860', 'ENSG00000172260', 'ENSG00000141424',
      'ENSG00000164512',
      ...,
      'ENSG00000164114', 'ENSG00000151702', 'ENSG00000224578',
      'ENSG00000138756', 'ENSG00000111052', 'ENSG00000176946',
      'ENSG00000150456', 'ENSG00000284934', 'ENSG00000261842',
      'ENSG00000260456'],
      dtype='object', length=17374)
```

```
[ ]: dc.mt.ulm(data=bm_mtx,
      net=markers,
      tmin = 3)
```

```
[ ]: score = dc.pp.get_obs(bm_mtx, key="score_ulm")
```

```
[ ]: bm_mtx.obs["score_ulm"].head(1)
```

```
[ ]:
      Acinar cells  Adipocytes  Adrenergic neurons  \
index
Guo-AAACCTGAGAGCTTCT-2      0.883366      0.01644      -0.592726

      Airway goblet cells  Alpha cells  \
index
Guo-AAACCTGAGAGCTTCT-2      -0.662709      0.83584

      Alveolar macrophages  Astrocytes  B cells  \
index
Guo-AAACCTGAGAGCTTCT-2      0.855729      0.297566  1.485696

      B cells memory  B cells naive  ...  \
index
Guo-AAACCTGAGAGCTTCT-2      -0.306571      0.23799  ...

      T follicular helper cells  T helper cells  \
index
Guo-AAACCTGAGAGCTTCT-2      -0.5133      -0.422886

      T memory cells  T regulatory cells  Tanycytes  \
index
Guo-AAACCTGAGAGCTTCT-2      1.642851      -0.004406  0.544296

      Taste receptor cells  Thymocytes  Trophoblast cells  \
index
Guo-AAACCTGAGAGCTTCT-2      0.413771  -0.543441      -0.5133

      Tuft cells  Urothelial cells
index
Guo-AAACCTGAGAGCTTCT-2      3.11394      -0.784177

[1 rows x 126 columns]
```

```
[ ]: bm_mtx.obsm["score_ulm"].columns
```

```
[ ]: Index(['Acinar cells', 'Adipocytes', 'Adrenergic neurons',
      'Airway goblet cells', 'Alpha cells', 'Alveolar macrophages',
      'Astrocytes', 'B cells', 'B cells memory', 'B cells naive',
      ...,
      'T follicular helper cells', 'T helper cells', 'T memory cells',
      'T regulatory cells', 'Tanycytes', 'Taste receptor cells', 'Thymocytes',
      'Trophoblast cells', 'Tuft cells', 'Urothelial cells'],
      dtype='object', length=126)
```

```
[ ]: #rank genes
```

```
bm_gene_rank = dc.tl.rankby_group(score, groupby="leiden_res_",
    ↪reference="rest", method="t-test_overestim_var")
bm_gene_rank = bm_gene_rank[bm_gene_rank["stat"] > 0]
bm_gene_rank.head(5)
```

```
[ ]:  group reference      name      stat  meanchange  pval  padj
      0      0      rest  Neutrophils  114.411587   6.383910  0.0   0.0
      1      0      rest  Macrophages   88.435972   5.630205  0.0   0.0
      2      0      rest  Dendritic cells  84.675829   5.373598  0.0   0.0
      3      0      rest   Microglia   83.824771   4.274959  0.0   0.0
      4      0      rest   Monocytes   80.601857   5.561706  0.0   0.0
```

```
[ ]: top_names_per_group = bm_gene_rank.groupby('group')['name'].apply(lambda x: x.
    ↪head(1))
display(top_names_per_group)
```

/tmp/ipython-input-2279155867.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

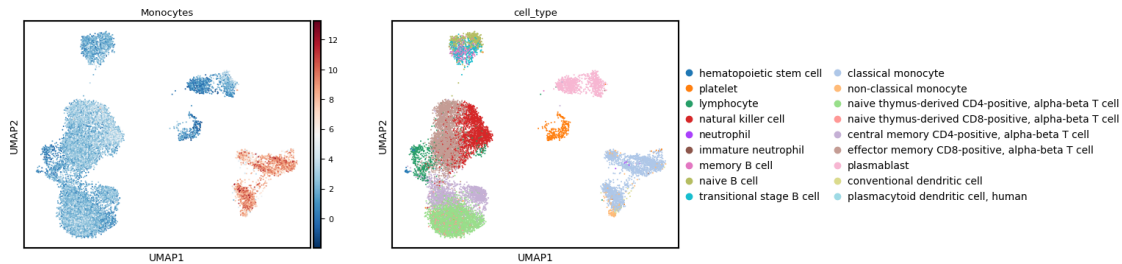
```
top_names_per_group = bm_gene_rank.groupby('group')['name'].apply(lambda x:
x.head(1))
```

```
group
0      0      Neutrophils
1     126      NK cells
2     252      Nuocytes
3     378      NK cells
4     504  B cells naive
5     630      Platelets
6     756      Nuocytes
7     882      Plasma cells
8    1023  Gamma delta T cells
9    1134      Monocytes
```

Name: name, dtype: category

Categories (126, object): ['Acinar cells', 'Adipocytes', 'Adrenergic neurons',
 'Airway goblet cells', ..., 'Thymocytes', 'Trophoblast
 ↪cells', 'Tuft cells',
 'Urothelial cells']

```
[ ]: sc.pl.umap(score, color=["Monocytes", "cell_type"], cmap="RdBu_r")
```

```
[ ]: n_ctypes = 1
      ctypes_dict = bm_gene_rank.groupby("group").head(n_ctypes).
      ↪groupby("group")["name"].apply(lambda x: list(x)).to_dict()
      ctypes_dict
```

/tmp/ipython-input-3463462141.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
      ctypes_dict = bm_gene_rank.groupby("group").head(n_ctypes).groupby("group")["name"]
      ↪.apply(lambda x: list(x)).to_dict()
```

```
[ ]: {'0': ['Neutrophils'],
      '1': ['NK cells'],
      '2': ['Nuocytes'],
      '3': ['NK cells'],
      '4': ['B cells naive'],
      '5': ['Platelets'],
      '6': ['Nuocytes'],
      '7': ['Plasma cells'],
      '8': ['Gamma delta T cells'],
      '9': ['Monocytes']}
```

```
[ ]: dict_ann = bm_gene_rank[bm_gene_rank["stat"] > 0].groupby("group").head(1).
      ↪set_index("group")["name"].to_dict()
      dict_ann
```

/tmp/ipython-input-4239545895.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.

```
      dict_ann = bm_gene_rank[bm_gene_rank["stat"] >
      0].groupby("group").head(1).set_index("group")["name"].to_dict()
```

```
[ ]: {'0': 'Neutrophils',
      '1': 'NK cells',
      '2': 'Nuocytes',
```

```
'3': 'NK cells',
'4': 'B cells naive',
'5': 'Platelets',
'6': 'Nuocytes',
'7': 'Plasma cells',
'8': 'Gamma delta T cells',
'9': 'Monocytes'}
```

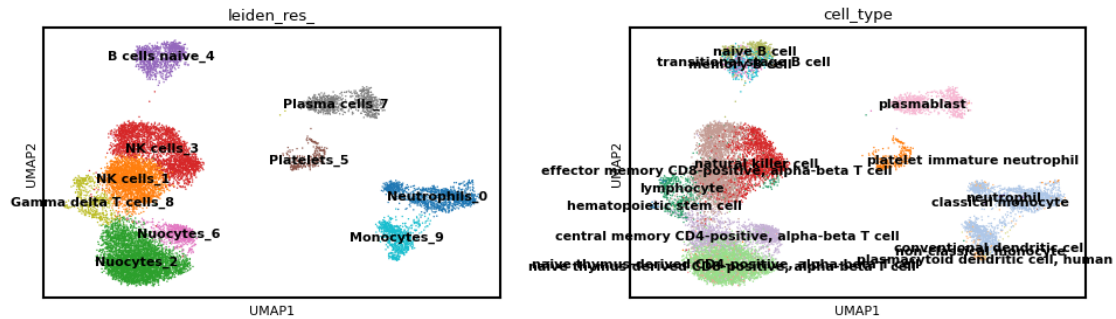
```
[ ]: dict_ann_unique = {k: v + '_' + str(k) for k, v in dict_ann.items()}
display(dict_ann_unique)
```

```
{'0': 'Neutrophils_0',
'1': 'NK cells_1',
'2': 'Nuocytes_2',
'3': 'NK cells_3',
'4': 'B cells naive_4',
'5': 'Platelets_5',
'6': 'Nuocytes_6',
'7': 'Plasma cells_7',
'8': 'Gamma delta T cells_8',
'9': 'Monocytes_9'}
```

```
[ ]: bm_mtx.obs["leiden_res_"] = bm_mtx.obs["leiden_res_"].cat.
      ↪ rename_categories(dict_ann_unique)
```

```
[ ]: plt.rcParams["figure.figsize"] = (5,3) # Adjust figure size
      # Reduce the default font size for text elements in plots
plt.rcParams['font.size'] = 8
plt.rcParams['axes.labelsize'] = 8
plt.rcParams['xtick.labelsize'] = 8
plt.rcParams['ytick.labelsize'] = 8

# Now replot the UMAP
sc.pl.umap(
    adata=bm_mtx,
    color=[ "leiden_res_", 'cell_type'],
    ncols=2,
    legend_loc = 'on data',
    size=4,
)
```

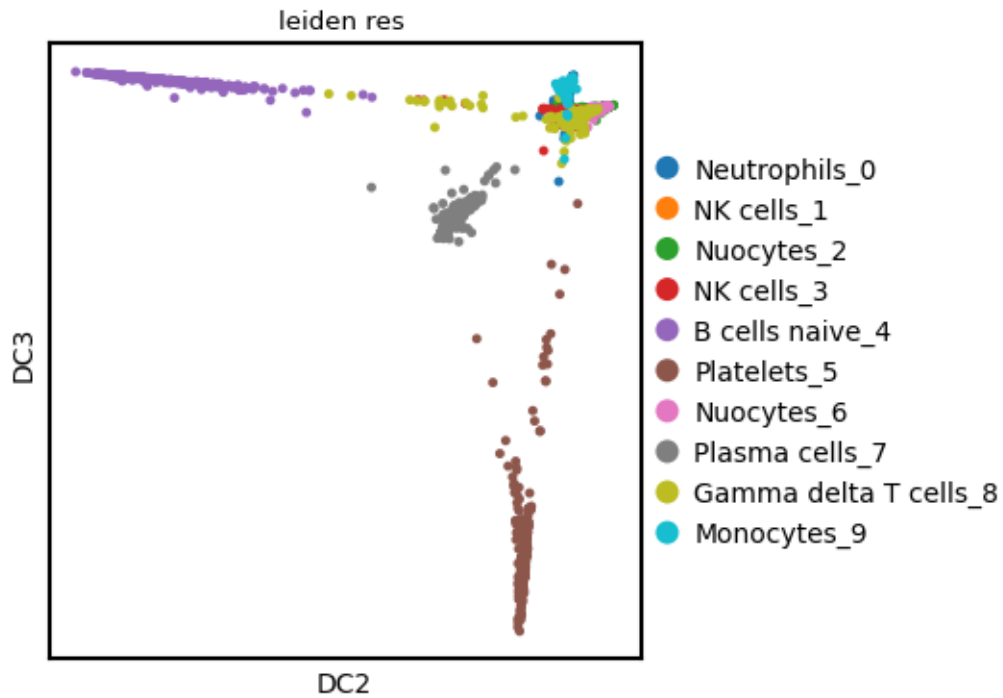


```
[ ]: #Pseudotime inference
sc.tl.diffmap(bm_mtx)
```

```
[ ]: plt.rcParams["figure.figsize"] = (4,4)
plt.rcParams['axes.labelsize'] = 10

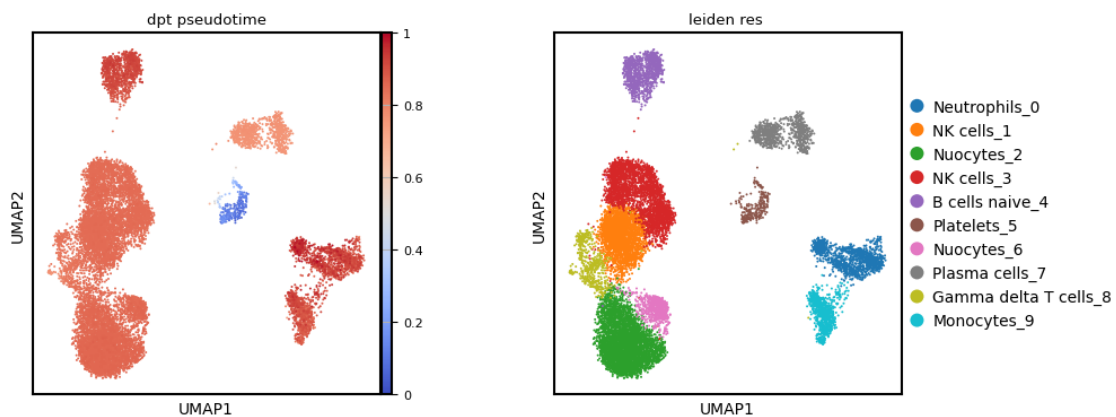
root_ixs = bm_mtx.obsm["X_diffmap"][:, 3].argmin()
sc.pl.scatter(
    bm_mtx,
    basis="diffmap",
    color=["leiden_res_"],
    components=[2, 3],
    size=50
)

bm_mtx.uns["iroot"] = root_ixs
```



```
[ ]: sc.tl.dpt(bm_mtx)
```

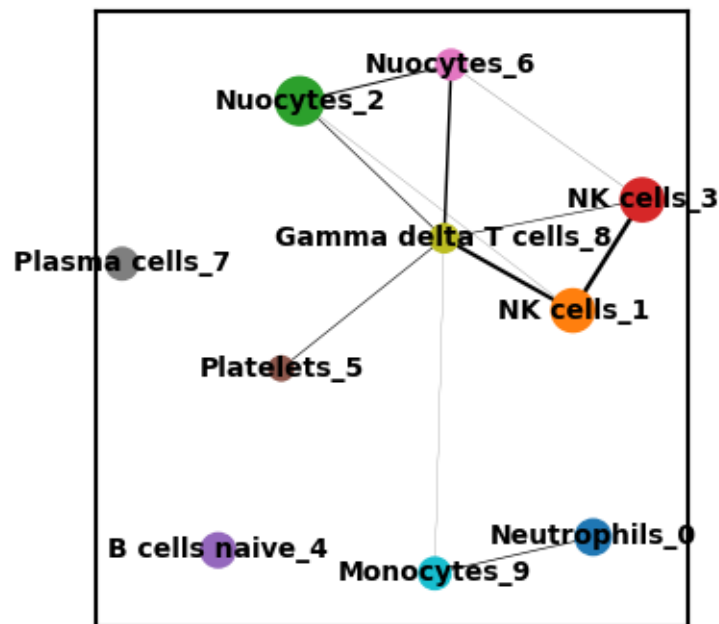
```
[ ]: sc.pl.scatter(
    bm_mtx,
    basis="umap",
    color=["dpt_pseudotime", 'leiden_res_'],
    color_map="coolwarm",
)
```

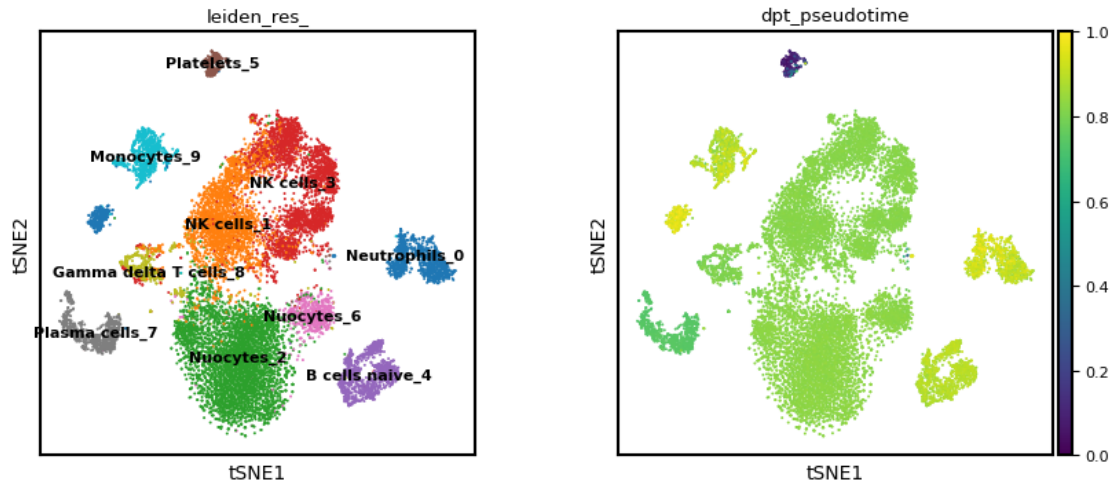


```
[ ]: #Trajectory analysis
root_celltype = "Platelets_5"

sc.pp.neighbors(bm_mtx, use_rep="X_diffmap")
sc.tl.paga(bm_mtx, groups="leiden_res_")
iroot = np.flatnonzero(bm_mtx.obs["leiden_res_"] == root_celltype)[0]
bm_mtx.uns["iroot"] = iroot
sc.tl.dpt(bm_mtx)

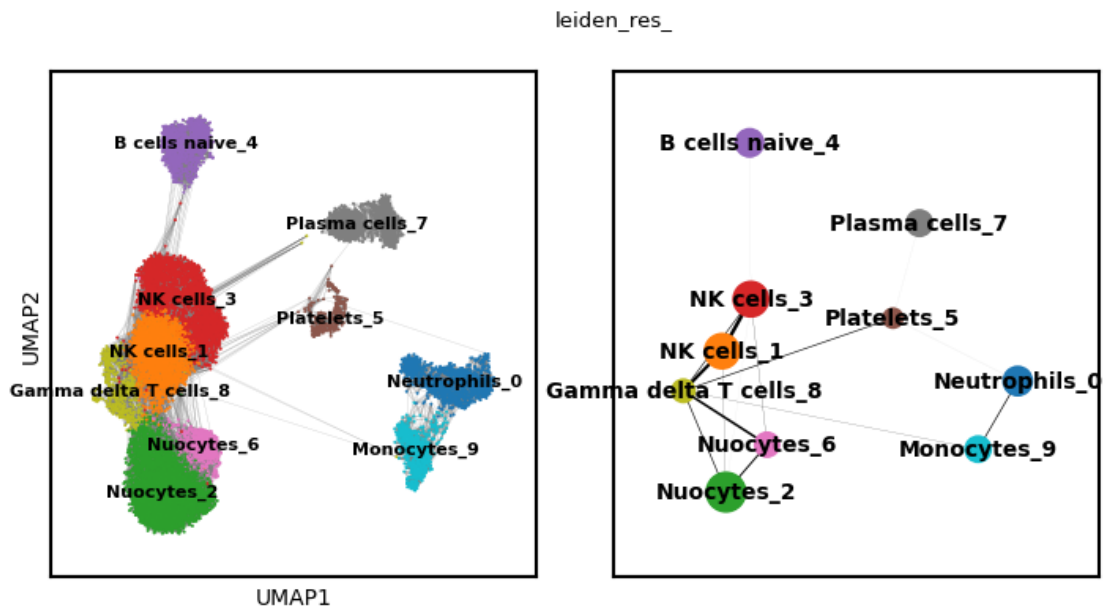
[ ]: sc.pl.paga(bm_mtx, color=["leiden_res_"])
sc.pl.tsne(bm_mtx, color=["leiden_res_", "dpt_pseudotime"], legend_loc="on_
↳data")
```





```
[ ]:
```

```
[ ]: sc.pl.paga_compare(bm_mtx, threshold=0.001, frameon=True, edges=True)
```



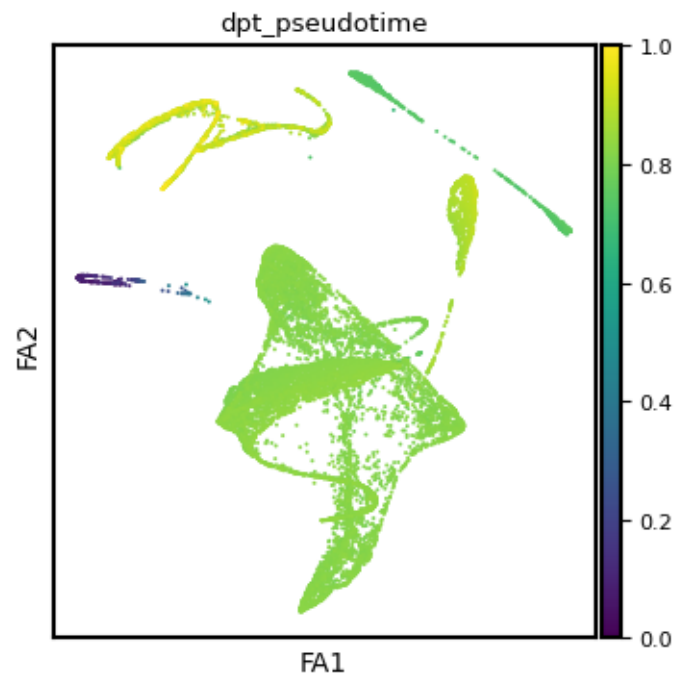
```
[ ]: [<Axes: xlabel='UMAP1', ylabel='UMAP2'>, <Axes: >]
```

```
[ ]: bm_mtx.uns['iroot'] = np.flatnonzero(bm_mtx.obs['leiden_res_'] ==  
    ↪ 'Platelets_5')[0]  
sc.tl.dpt(bm_mtx)
```

```
[ ]: bm_mtx
```

```
[ ]: AnnData object with n_obs × n_vars = 14783 × 17374
      obs: 'disease stage', 'treatment', 'timepoint', 'Dataset', 'sample',
          'disease_original', 'disease_general', 'COVID-19 Condition', 'Lineage',
          'Cell.group', 'Cell.class_reannotated', 'n_genes', 'n_counts', 'percent_mito',
          'tissue_original', 'tissue_ontology_term_id', 'disease_ontology_term_id',
          'donor_id', 'development_stage_ontology_term_id', 'assay_ontology_term_id',
          'cell_type_ontology_term_id', 'self_reported_ethnicity_ontology_term_id',
          'sex_ontology_term_id', 'is_primary_data', 'suspension_type', 'tissue_type',
          'cell_type', 'assay', 'disease', 'sex', 'tissue', 'self_reported_ethnicity',
          'development_stage', 'observation_joinid', 'n_genes_by_counts',
          'log1p_n_genes_by_counts', 'total_counts', 'log1p_total_counts',
          'pct_counts_in_top_50_genes', 'pct_counts_in_top_100_genes',
          'pct_counts_in_top_200_genes', 'pct_counts_in_top_500_genes', 'total_counts_MT',
          'log1p_total_counts_MT', 'pct_counts_MT', 'total_counts_RIBO',
          'log1p_total_counts_RIBO', 'pct_counts_RIBO', 'total_counts_HB',
          'log1p_total_counts_HB', 'pct_counts_HB', 'leiden_res_', 'dpt_pseudotime'
      var: 'n_cells', 'feature_is_filtered', 'feature_name', 'feature_reference',
          'feature_biotype', 'feature_length', 'feature_type', 'MT', 'RIBO', 'HB',
          'n_cells_by_counts', 'mean_counts', 'log1p_mean_counts',
          'pct_dropout_by_counts', 'total_counts', 'log1p_total_counts',
          'highly_variable', 'means', 'dispersions', 'dispersions_norm'
      uns: 'citation', 'doi', 'organism', 'organism_ontology_term_id',
          'schema_reference', 'schema_version', 'title', 'cell_type_colors', 'log1p',
          'hvg', 'pca', 'neighbors', 'umap', 'leiden_res_', 'leiden_res__colors',
          'diffmap_evals', 'iroot', 'paga', 'leiden_res__sizes'
      obsm: 'X_pca', 'X_tsne', 'X_umap', 'score_ulm', 'padj_ulm', 'X_diffmap'
      varm: 'PCs'
      layers: 'counts'
      obsp: 'distances', 'connectivities'
```

```
[ ]: sc.tl.draw_graph(bm_mtx, layout='fa')
      sc.pl.draw_graph(bm_mtx, color=['dpt_pseudotime'], legend_loc='on data')
```



```
[ ]:
```